Does a higher minimum wage cost jobs?

Card and Krueger (1994), explained in eight slides

Example output for Round 4

Made by Claude Code with the Round 4 slide-deck prompt

The question

- The textbook competitive model: make labour more expensive and employers hire less.
- Fast food is the test case: many workers earn at or near the minimum.
- Most earlier studies compared periods over time, where many things change at once.

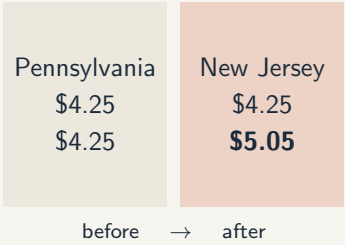
The idea

Find two places that are alike, raise the minimum wage in only one of them, and compare.

Source: paper, Sections I–II

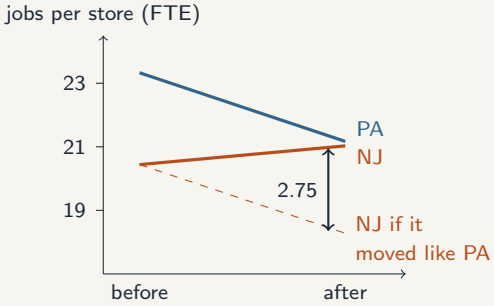
The natural experiment

- 1 April 1992: New Jersey raises its minimum from \$4.25 to \$5.05.
- Eastern Pennsylvania keeps \$4.25.
- 410 Burger King, KFC, Roy Rogers and Wendy's restaurants surveyed by telephone.
- Before: February–March 1992. After: seven to eight months later.



Source: paper, Sections I–II

Difference in differences



Pennsylvania lost about two jobs per store; New Jersey gained a little.

If New Jersey would otherwise have moved like Pennsylvania, the minimum wage is associated with about 2.75 more jobs per store, roughly 13 percent.

Source: our replication of Table 3; 13 percent from the paper's text

The finding

Comparison	Estimate	Std. error
New Jersey vs Pennsylvania, 357 stores	+2.33 jobs	1.19
Same, with chain and ownership controls	+2.30 jobs	1.20
Wage gap measure (stores that needed the largest raise)	+15.65	6.08

In words

No sign that employment fell. If anything it rose where the minimum wage bit hardest. Meal prices rose in New Jersey relative to Pennsylvania.

Source: Table 4 of the paper, reproduced exactly in our replication

How robust is it?

Choice	What happens to the New Jersey estimate
Which stores	384 instead of 357 stores: 2.75 instead of 2.33
Part-time weight	0 to 1 instead of 0.5: between 2.55 and 2.10
Closed stores	Dropping the six that closed: 2.28
Standard errors	Robust instead of conventional: 1.45 instead of 1.19, so t falls from about 1.96 to 1.61
Proportional change	+5.2 percent, standard error 5.1 percent

The sign never flips; the statistical significance depends on choices.

Source: our replication

What the study cannot show

- **Two states.** A shock that hit only one of them looks like a minimum-wage effect.
- **Eight months.** Long-run responses such as fewer new restaurants or more automation are outside the window.
- **Fast food only.** Other low-wage sectors may respond differently.
- **Survey answers.** Repeat interviews of the same stores agreed only moderately on employment (reliability 0.70).

Source: paper; the list of limits is our reading

Why it mattered

- It challenged a textbook prediction with a simple, transparent design.
- It started a long debate: a critique using payroll records (Neumark and Wascher, 2000) and a reply using government employment data (Card and Krueger, 2000).
- The design, comparing a treated and an untreated group before and after, became a standard tool in economics.
- David Card shared the 2021 Nobel memorial prize in economics, in part for this line of work.

Source: background knowledge of the assistant; check before teaching

A higher minimum wage did not visibly cost jobs in New Jersey fast food in 1992.

Whether that holds elsewhere is a question of design, not of software.